Plotting coordinates generated from a rule

1 A rule for a set of coordinates can be written as an __________. Tick **1** correct answer

- [ ] addition
- [x] equation
- [ ] exponential
- [ ] identity
- [ ] origin

2 Match the equation to the coordinates that follow the rule. Write the correct letter in each box

a	$x = 9$
b	$y = 9$
c	$y = 3x$
d	$y = x + 9$
e	$y = x - 9$
f	$y = -3x$

f	$(3, -9)$ $(0, 0)$ $(5, -15)$ $(-3, 9)$
e	$(9, 0)$ $(3, -6)$ $(0, -9)$ $(-9, -18)$
a	$(9, 0)$ $(9, 9)$ $(9, -9)$ $(9, 3)$
b	$(9, 9)$ $(3, 9)$ $(-3, 9)$ $(0, 9)$
c	$(-3, -9)$ $(0, 0)$ $(-1, -3)$ $(3, 9)$
d	$(0, 9)$ $(-9, 0)$ $(3, 12)$ $(-3, 6)$

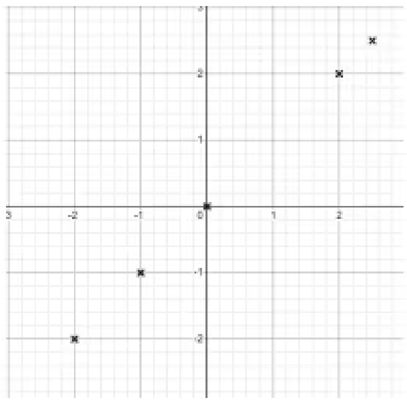
3 What is the rule for the coordinates $(0, 0)$ $(5, 1)$ $(-10, -2)$ $(15, 3)$? Tick **2** correct answers

- [ ] The y coordinate equals the x coordinate minus 4
- [ ] The y coordinate equals the x coordinate multiplied by 5
- [x] The y coordinate equals the x coordinate divided by 5
- [x] The x coordinate equals the y coordinate multiplied by 5
- [ ] The x coordinate equals the y coordinate plus 4

4 Fill in the missing value in the following coordinate so it follows the rule: $y = x + 3.5$

(3.2 , 6.7) Fill in the blank

5 Which of these rules match the graph shown? Tick **1** correct answer



- [ ] $y = 2$
- [ ] $y = -x$
- [ ] $y = 2x$
- [x] $y = x$

6 Which set of coordinates will make a straight line when plotted? Tick **1** correct answer

- [ ] $(10, 5)$ $(8, 3)$ $(0, 5)$ $(4, 8)$
- [ ] $(0, 0)$ $(1, 3)$ $(-1, 3)$ $(6, 2)$
- [ ] $(1, 1)$ $(-6.2, 6.2)$ $(-7.1, -7.1)$ $(2.5, -2.5)$
- [x] $(0, 0)$ $(5, 2.5)$ $(-10, -5)$ $(8, 4)$